

#W[1] = FPT: Fixed-Parameter Tractable Exact Algorithms for the # k -Matching Problem

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The concept of NP-completeness has been proposed for half a century, and it is conjectured that there are no subexponential-time algorithms for NP-hard problems, which is known as the Exponential Time Hypothesis (ETH). As a pivotal conjecture in the field of theoretical computer science, numerous conjectures in computer science rely on ETH. A corollary of the Exponential Time Hypothesis is the Counting Exponential Time Hypothesis (#ETH), and a further corollary of #ETH is that #W[1] $\neq$ FPT. The # k -matching problem is a well-known #W[1]-complete problem. We have discovered an algorithm for the # k -matching problem with a running time of $f(k)n^{O(1)}$. This result implies that the hypotheses #W[1] $\neq$ FPT, W[1] $\neq$ FPT, the Counting Exponential Time Hypothesis, and the Exponential Time Hypothesis all do not hold.

Additional Key Words and Phrases: Computational Complexity, Parameterized Complexity, # k -Matching, Exponential Time Hypothesis, Fixed-Parameter Tractability

1 Introduction

Since the formal proposal of the concept of NP-completeness, the field of theoretical computer science has engaged in half a century of in-depth exploration and intense debate on the computational complexity of such problems [2]. The core challenge of NP-hard problems is that no polynomial-time algorithm has been found for their solution to date, and it is widely conjectured that there exist no efficient subexponential-time algorithms for these problems. This conjecture has been condensed into the highly influential Exponential Time Hypothesis (ETH) in theoretical computer science [5]. As a foundational conjecture in the complexity theory framework, ETH not only profoundly characterizes the computational lower bounds of NP-hard problems but also serves as the logical premise for numerous important computational conjectures and theoretical deductions. Verifying its rationality and validity has long been a research hotspot and core topic in the field.

The value of ETH lies not only in its characterization of NP-hard problems themselves but also in its ability to derive a series of far-reaching complexity conclusions, including a key corollary in the field of parameterized complexity: #W[1] $\neq$ FPT. Parameterized complexity is a refined complexity analysis framework for intractable problems, whose core is to decompose the input of a problem into two parts—the *main input* (with size denoted by n) and the *parameter* (denoted by k)—and characterize the impact of the parameter on the time complexity of an algorithm independently. A Fixed-Parameter Tractable (FPT) algorithm is the core criterion for efficient algorithms in this framework, with a strict definition: a problem is fixed-parameter tractable (FPT) if there exists a computable function $f(k)$ that depends only on the parameter k , such that the algorithm can solve the problem in $f(k) \cdot n^{O(1)}$ time. Here, $f(k)$ is typically an exponential or higher-order function of k , and $n^{O(1)}$ denotes polynomial time with respect to the main input size n . This complexity form implies that when the parameter k is much smaller than the input size n , the algorithm can handle a main input of any size in polynomial time of n . The relationship between #W[1] and FPT directly determines whether a class of counting problems admits parameterized efficient algorithms, serving as an important link connecting classical complexity and parameterized complexity. ETH naturally implies the Counting Exponential Time Hypothesis (#ETH), and it has been proven that #W[1] $\neq$ FPT under #ETH [4], meaning that no FPT algorithms exist for #W[1]-hard problems. In addition, W[1] $\neq$ FPT is another important conjecture in parameterized complexity [1], from which #W[1] $\neq$ FPT can be directly deduced.

As a classic problem in counting complexity, the $\#k$ -matching problem has been proven to be $\#W[1]$ -hard. By means of an extremely concise elementary algebraic method, we present an algorithm for the $\#k$ -matching problem with a running time of $f(k)n^{O(1)}$. This result implies that the long-standing conjectures of $\#W[1] \neq \text{FPT}$, $W[1] \neq \text{FPT}$, the Counting Exponential Time Hypothesis, and the Exponential Time Hypothesis all do not hold. In addition, as the strengthened versions of the Exponential Time Hypothesis and the Counting Exponential Time Hypothesis, the Strong Exponential Time Hypothesis[5, 6] and the Counting Strong Exponential Time Hypothesis[7] do not hold either.

2 Method

We study the k -matching counting problem. Let $G = (V, E)$ denote the host graph for which we count k -matchings, A be the adjacency matrix of G , $A_{[i,j]}$ represent the (i, j) -th entry of A , $n = |V|$ be the number of vertices in G , $N(k)$ be the number of k -matchings in G , and $P(n, k)$ be the set of k -permutations of $\{1, 2, \dots, n\}$.

The set $P(n, n)$ contains $n!$ permutations, denoted as $\phi_1, \phi_2, \dots, \phi_{n!}$, where $\phi(j)$ denotes the j -th element of the n -permutation ϕ . We denote $n!$ subgraphs of G as $G_1 = (V_1, E_1), G_2 = (V_2, E_2), \dots, G_{n!} = (V_{n!}, E_{n!})$, where for all $1 \leq i \leq n!$, $V_i = V$ and $E_i = \{(a, b) \mid (a, b) \in E \wedge b = \phi_i(a)\}$.

Since each vertex j in G_i has at most one out-edge $(j, \phi_i(j))$, the number of k -directed matchings in G_i can be written as $\frac{1}{k!} \sum_{\sigma \in P(n, k)} \prod_{j=1}^k A_{[\sigma(j), \phi_i(\sigma(j))]}$. Here, a k -directed matching is defined as a set of k directed edges passing through $2k$ distinct vertices by definition.

Any k -directed matching $\{< i_1, j_1 >, < i_2, j_2 >, \dots, < i_k, j_k >\}$ in G is a k -directed matching in G_i if and only if $\phi_i(i_h) = j_h$ for all $1 \leq h \leq k$. There are exactly $(n - k)!$ such permutations in $P(n, n)$. Meanwhile, one k -matching corresponds to 2^k k -directed matchings. Thus, we have:

Lemma 1. $N(k) = \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{i=1}^{n!} \sum_{\sigma \in P(n, k)} \prod_{j=1}^k A_{[\sigma(j), \phi_i(\sigma(j))]}$.

The article should be revised.

Theorem 1. $N(k) = \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{\sigma \in P(n, k)} \sum_{\phi \in P(n, k)} \prod_{j=1}^k A_{[\sigma(j), \phi(j)]}$.

Proof. For $\forall \phi \in P(n, k)$, we denote the subpermutation $\phi(j_1), \phi(j_2), \dots, \phi(j_l)$ of ϕ as $\phi(j_1, j_2, \dots, j_l)$, and the subpermutation of ϕ obtained by removing $\phi(j_1), \phi(j_2), \dots, \phi(j_l)$ as $\bar{\phi}(j_1, j_2, \dots, j_l)$. Let $P(n, \bar{\phi}(j_1, j_2, \dots, j_l))$ be the set of $(n - l)$ -permutations of the set $\{a \mid 1 \leq a \leq n \wedge \forall 1 \leq i \leq l, a \neq \phi(j_i)\}$.

From Theorem 1:

$$\begin{aligned}
N(k) &= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{i=1}^{n!} \sum_{\sigma \in P(n,k)} \prod_{j=1}^k A_{[\sigma(j), \phi(\sigma(j))]} \\
&= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{\sigma \in P(n,k)} \sum_{i=1}^{n!} \prod_{j=1}^k A_{[\sigma(j), \phi(\sigma(j))]} \\
&= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{\sigma \in P(n,k)} \sum_{\phi \in P(n,n)} \prod_{j=1}^k A_{[\sigma(j), \phi(\sigma(j))]} \\
&= \frac{1}{k!} \cdot \frac{1}{(n-k)!} \cdot \frac{1}{2^k} \sum_{\sigma \in P(n,k)} \sum_{\substack{\phi(\sigma(1), \sigma(2), \dots, \sigma(k)) \\ \in P(n,k)}} \sum_{\substack{\tilde{\phi}(\sigma(1), \sigma(2), \dots, \sigma(k)) \\ \in P(n, \tilde{\phi}(\sigma(1), \sigma(2), \dots, \sigma(k)))}} \prod_{j=1}^k A_{[\sigma(j), \phi(\sigma(j))]} \\
&= \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{\sigma \in P(n,k)} \sum_{\substack{\phi(\sigma(1), \sigma(2), \dots, \sigma(k)) \\ \in P(n,k)}} \prod_{j=1}^k A_{[\sigma(j), \phi(\sigma(j))]} \\
&= \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{\sigma \in P(n,k)} \sum_{\phi \in P(n,k)} \prod_{j=1}^k A_{[\sigma(j), \phi(j)]}.
\end{aligned} \tag{1}$$

Lemmas 2 and 3 are trivial:

Lemma 2. $\sum_{\sigma \in P(n,2)} x_{\sigma(1)} x_{\sigma(2)} = \sum_{\sigma(1)=1}^n \sum_{\sigma(2)=1}^n x_{\sigma(1)} x_{\sigma(2)} - \sum_{\sigma(1)=1}^n x_{\sigma(1)} x_{\sigma(1)}.$

Lemma 3. If $\forall \leq i \leq n, x_i \in \{0, 1\}$, then $\sum_{\sigma \in P(n,2)} x_{\sigma(1)} x_{\sigma(2)} = \sum_{\sigma(1)=1}^n \sum_{\sigma(2)=1}^n x_{\sigma(1)} x_{\sigma(2)} - \sum_{\sigma(1)=1}^n x_{\sigma(1)}.$

It is not difficult to derive from Lemma 3 that:

Lemma 4. If $\forall 1 \leq i \leq n, x_i \in \{0, 1\}$, then $\forall k > 1$, there exist $g'_k(1), g'_k(2), \dots, g'_k(k)$ such that

$$\sum_{\sigma \in P(n,k)} \prod_{i=1}^k x_{\sigma(i)} = \sum_{l=1}^k g'_k(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} = \sum_{l=1}^k g'_k(l) \prod_{h=1}^l \sum_{1 \leq i_h \leq n} x_{i_h}.$$

Proof. We prove this by mathematical induction:

Base case: Already established by Lemma 3.

Inductive step: The inductive hypothesis is

$$\sum_{\sigma \in P(n,k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} = \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h}. \tag{2}$$

The term $\sum_{\sigma \in P(n,k)} \prod_{i=1}^k x_{\sigma(i)}$ can be rewritten as

$$\begin{aligned}
\sum_{\substack{\sigma(k)=1 \\ \sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n,k-1) \\ \sigma(1), \sigma(2), \dots, \sigma(k-1) \neq \sigma(k)}}^n \sum_{\sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n,k-1)} \prod_{i=1}^k x_{\sigma(i)} &= \sum_{i=1}^n x_i \sum_{\sigma \in P(n,k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} - \sum_{j=1}^{k-1} \sum_{\sigma \in P(n,k-1)} x_{\sigma(j)} \prod_{i=1}^{k-1} x_{\sigma(i)} \\
&= \sum_{i=1}^n x_i \sum_{\sigma \in P(n,k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} - (k-1) \sum_{\sigma \in P(n,k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)}.
\end{aligned} \tag{3}$$

From the inductive hypothesis:

$$\begin{aligned}
\sum_{i=1}^n x_i \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} &= \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \sum_{i=1}^n x_i \prod_{h=1}^l x_{i_h} \\
&= \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} x_{i_{l+1}} \prod_{h=1}^l x_{i_h} \\
&= \sum_{l=1}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_{l+1} \leq n} \prod_{h=1}^{l+1} x_{i_h}.
\end{aligned} \tag{4}$$

Substituting (1) and (3) into (2), we obtain:

$$\begin{aligned}
\sum_{\sigma \in P(n, k)} \prod_{i=1}^k x_{\sigma(i)} &= \sum_{i=1}^n x_i \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} - (k-1) \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{\sigma(i)} \\
&= g'_{k-1}(k-1) \sum_{1 \leq i_1, i_2, \dots, i_k \leq n} \prod_{h=1}^k x_{i_h} \\
&\quad + \sum_{l=2}^{k-1} g'_{k-1}(l-1) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} \\
&\quad - (k-1) \sum_{l=2}^{k-1} g'_{k-1}(l) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} - (k-1) g'_{k-1}(1) \sum_{1 \leq i_1 \leq n} x_{i_1} \\
&= \sum_{l=2}^{k-1} (g'_{k-1}(l-1) - (k-1) g'_{k-1}(l)) \sum_{1 \leq i_1, i_2, \dots, i_l \leq n} \prod_{h=1}^l x_{i_h} \\
&\quad + g'_{k-1}(k-1) \sum_{1 \leq i_1, i_2, \dots, i_k \leq n} \prod_{h=1}^k x_{i_h} - (k-1) g'_{k-1}(1) \sum_{1 \leq i_1 \leq n} x_{i_1}.
\end{aligned} \tag{5}$$

Let

$$g'_k(l) = \begin{cases} g'_{k-1}(k-1), & l = k, \\ g'_{k-1}(l-1) - (k-1) g'_{k-1}(l), & 1 < l < k, \\ -(k-1) g'_{k-1}(1), & l = 1. \end{cases}$$

Furthermore, let

$$g'_2(l) = 1 \quad \forall 1 \leq l \leq 2.$$

This completes the inductive step.

Clearly, we only need $O(k^3)$ time to compute $g_k(l)$ according to the recursive formula in the proof of Lemma 4.

It follows from Lemma 4 that:

Theorem 2. $N(k) = \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{l=1}^k \frac{(n-l)!}{(k-l)!} g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n, l)} \prod_{h=1}^l A_{[j_h, \phi(h)]}.$

Proof. From Theorem 1:

$$N(k) = \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{\phi \in P(n, k)} \sum_{\sigma \in P(n, k)} \prod_{j=1}^k A_{[\sigma(j), \phi(j)]}. \tag{6}$$

From Lemma 4:

$$\sum_{\sigma \in P(n,k)} \prod_{j=1}^k A_{[\sigma(j), \phi(j)]} = \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l A_{[j_h, \phi(h)]}. \quad (7)$$

Substituting (7) into (6), we obtain:

$$\begin{aligned} N(k) &= \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n,k)} \prod_{h=1}^l A_{[j_h, \phi(h)]} \\ &= \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{l=1}^k |P(n-l, k-l)| g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n,l)} \prod_{h=1}^l A_{[j_h, \phi(h)]} \\ &= \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{l=1}^k |P(n-l, k-l)| g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n,l)} \prod_{h=1}^l A_{[j_h, \phi(h)]} \\ &= \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{l=1}^k \frac{(n-l)!}{(k-l)!} g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{\phi \in P(n,l)} \prod_{h=1}^l A_{[j_h, \phi(h)]}. \end{aligned} \quad (8)$$

It is evident that if we can compute $\sum_{1 \leq j_h \leq n} \sum_{\phi \in P(n,l)} \prod_{h=1}^l A_{[j_h, \phi(h)]}$ in fixed-parameter tractable (FPT) time, then $N(k)$ can be computed in FPT time as well.

We denote $\Pi(n, k)$ as the set of all unordered k -partitions of the set $\{1, 2, \dots, n\}$. That is, if $\{J_1, J_2, \dots, J_k\} \in \Pi(n, k)$, then $J_1 \cup J_2 \cup \dots \cup J_k = \{1, 2, \dots, n\}$, $\forall 1 \leq i_1 \neq i_2 \leq l, J_{i_1} \cap J_{i_2} = \emptyset$, and $\forall 1 \leq i \leq l, J_i \neq \emptyset$. Furthermore, we denote $S(n, k) = |\Pi(n, k)|$ (the Stirling numbers of the second kind) and $B_n = \sum_{k=1}^n S(n, k)$ (the n -th Bell number).

From the literature, we have:

Lemma 5 [3]. $B_n = n^{O(n)}$.

Lemma 6. $\forall k > 1$, there exist functions $f_1^k, f_2^k, \dots, f_k^k$ defined on $\Pi(k, 1), \Pi(k, 2), \dots, \Pi(k, k)$ respectively, such that

$$\sum_{\sigma \in P(n,k)} \prod_{i=1}^k x_{i\sigma(i)} = \sum_{l=1}^k \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k,l)} f_l^k(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}.$$

Proof. We prove this lemma by mathematical induction. The base case is established by Lemma 2, and we proceed to the inductive step:

The inductive hypothesis is

$$\sum_{\sigma \in P(n,k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} = \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1,l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}. \quad (9)$$

Clearly,

$$\begin{aligned}
\sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{\sigma(k)=1}^n \sum_{\substack{\sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n, k-1) \\ \sigma(1), \sigma(2), \dots, \sigma(k-1) \neq \sigma(k)}} \prod_{i=1}^k x_{i\sigma(i)} \\
&= \sum_{\sigma(k)=1}^n \sum_{\sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n, k-1)} \prod_{i=1}^k x_{i\sigma(i)} \\
&\quad - \sum_{j=1}^{k-1} \sum_{\sigma(1), \sigma(2), \dots, \sigma(k-1) \in P(n, k-1)} x_{k\sigma(j)} \prod_{i=1}^{k-1} x_{i\sigma(i)} \\
&= \sum_{h_k=1}^n x_{kh_k} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} - \sum_{j=1}^{k-1} x_{k\sigma(j)} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)}.
\end{aligned} \tag{10}$$

From the inductive hypothesis:

$$\begin{aligned}
\sum_{h_k=1}^n x_{kh_k} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{h_k=1}^n x_{kh_k} \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h} \\
&= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l, j_{l+1} \leq n} x_{kj_{l+1}} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}.
\end{aligned} \tag{11}$$

Moreover,

$$\begin{aligned}
\sum_{j=1}^{k-1} x_{k\sigma(j)} \sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{j=1}^{k-1} x_{k\sigma(j)} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h} \\
&= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{h'=1}^l |J_{h'}| x_{k\sigma(j_{h'})} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}.
\end{aligned} \tag{12}$$

Substituting (11) and (12) into (10), we obtain:

$$\begin{aligned}
\sum_{\sigma \in P(n, k-1)} \prod_{i=1}^{k-1} x_{i\sigma(i)} &= \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l, j_{l+1} \leq n} x_{kj_{l+1}} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h} \\
&\quad - \sum_{l=1}^{k-1} \sum_{\{J_1, J_2, \dots, J_l\} \in \Pi(k-1, l)} f_l^{k-1}(\{J_1, J_2, \dots, J_l\}) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{h'=1}^l |J_{h'}| x_{k\sigma(j_{h'})} \prod_{h=1}^l \prod_{i \in J_h} x_{ij_h}.
\end{aligned} \tag{13}$$

For any partition $\{J_1, J_2, \dots, J_l\} \in \Pi(k, l)$ with $k \in J_l$, we define:

$$f_l^k(\{J_1, J_2, \dots, J_l\}) = \begin{cases} f_{l-1}^{k-1}(\{J_1, J_2, \dots, J_{l-1}\}), & |J_l| = 1, \\ -(|J_l| - 1) f_l^{k-1}(\{J_1, J_2, \dots, J_{l-1}, J_l \setminus \{k\}\}), & |J_l| > 1, \end{cases}$$

This completes the proof of the lemma.

Clearly, f_l^k can be computed in $k^{O(k)}$ time using the recursive formula in the proof (this requires evaluating all $\sum_{l \leq m \leq k} S(m, l) = \sum_{m=1}^k B_m$ partitions in $\Pi(m, l)$ for all $l \leq m \leq k$).

Lemma 7.

$$N(k) = \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{l=1}^k g'_k(l) \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \sum_{q=1}^k \sum_{\{P_1, P_2, \dots, P_q\} \in \Pi(k, q)} f_q^k(\{P_1, P_2, \dots, P_q\}) \sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]}.$$

Proof. This follows immediately by substituting Lemma 6 into Theorem 2 and renaming duplicate symbols.

Theorem 3.

$$N(k) = \frac{1}{k!} \cdot \frac{1}{2^k} \sum_{l=1}^k \frac{(n-l)!}{(k-l)!} g'_k(l) \sum_{q=1}^k \sum_{\{P_1, P_2, \dots, P_q\} \in \Pi(k, q)} f_q^k(\{P_1, P_2, \dots, P_q\}) \sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]}.$$

Proof. This follows immediately from Lemma 7 by interchanging the order of summation.

Theorem 4. The expression $\sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]}$ can be computed in $O(n^2)$ time.

Proof.

$$\begin{aligned} \sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]} &= \prod_{h=1}^q \sum_{1 \leq p_h \leq n} \prod_{i \in P_h} \sum_{1 \leq j_i \leq n} A_{[j_i, p_h]} \\ &= \prod_{h=1}^q \sum_{1 \leq p_h \leq n} d_{p_h}^{|P_h|}, \end{aligned} \tag{14}$$

where d_{p_h} denote the in-degree of vertex p_h . Computing $\prod_{h=1}^q \sum_{1 \leq p_h \leq n} d_{p_h}^{|P_h|}$ only requires calculating $\sum_{1 \leq p_1 \leq n} d_{p_1}^{|P_1|}$, $\sum_{1 \leq p_2 \leq n} d_{p_2}^{|P_2|}$, $\dots$, $\sum_{1 \leq p_q \leq n} d_{p_q}^{|P_q|}$, followed by $(q-1)$ multiplications. The total time complexity is $O(n^2)$.

Theorem 5. $N(k)$ can be computed in fixed-parameter tractable (FPT) time.

Proof. By Lemma 5, $\sum_{l=1}^k S(k, l) = k^{O(k)}$. The formula in Theorem 3 requires evaluating a total of $\sum_{l=1}^k \sum_{q=1}^k B_q$ expressions of the form

$$\sum_{1 \leq p_1, p_2, \dots, p_q \leq n} \sum_{1 \leq j_1, j_2, \dots, j_l \leq n} \prod_{h=1}^q \prod_{i \in P_h} A_{[j_i, p_h]},$$

with a time complexity of $k^{O(k)} O(n^2)$. Additionally, the time complexity of deriving the expression in Theorem 3 is $k^{O(k)}$. Thus, $N(k)$ is fixed-parameter tractable (FPT).

It follows directly from Theorem 5 that:

Theorem 6. $\#W[1] = \text{FPT}$.

Theorem 7. $W[1] = \text{FPT}$.

Theorem 8. The Counting Exponential Time Hypothesis (#ETH) does not hold.

Theorem 9. The Exponential Time Hypothesis (ETH) does not hold.

Theorem 10. The Counting Strong Exponential Time Hypothesis (#SETH) does not hold.

Theorem 11. The Strong Exponential Time Hypothesis (SETH) does not hold.

3 Conclusion

As a classic problem in the interdisciplinary field of counting complexity and parameterized complexity, the #k-matching problem has long been proven by the academic community to be a canonical #W[1]-hard problem, serving as a core benchmark for verifying the existence of parameterized efficient algorithms for such counting problems. We conduct algorithm design and derivation via algebraic methods, where the #k-matching problem refers to the counting problem of enumerating the number of edge-disjoint k-matchings in an undirected graph. Ultimately, we successfully

present an $f(k)n^{O(1)}$ -time algorithm for the $\#k$ -matching problem, whose complexity fully adheres to the strict definition of fixed-parameter tractability (FPT). This research result directly demonstrates that the long-standing hypotheses of $\#W[1] \neq \text{FPT}$ and $W[1] \neq \text{FPT}$, which have perplexed the field of theoretical computer science for decades, do not hold. Consequently, the Counting Exponential Time Hypothesis ($\#ETH$), the Counting Strong Exponential Time Hypothesis ($\#SETH$), the Strong Exponential Time Hypothesis (SETH) and the Exponential Time Hypothesis (ETH), as their theoretical cornerstones, are also falsified.

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Received 28 January 2026